

COMPARATIVE ANALYSIS

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Q1. Object Detection Performance Comparison

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A (Viola-Jones)	80	20	30
Model B (YOLO)	90	15	20

(a) Precision and Recall Calculation

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN)$$

Model A: Precision = $80 / (80 + 20) = 80 / 100 = 0.80 \rightarrow 80\%$

Model A: Recall = $80 / (80 + 30) = 80 / 110 = 0.727 \rightarrow 72.7\%$

Model B: Precision = $90 / (90 + 15) = 90 / 105 = 0.857 \rightarrow 85.7\%$

Model B: Recall = $90 / (90 + 20) = 90 / 110 = 0.818 \rightarrow 81.8\%$

Model	Precision	Recall
Model A (Viola-Jones)	80.0%	72.7%
Model B (YOLO)	85.7%	81.8%

(b) Comparison and Evaluation

Model B (YOLO) outperforms Model A (Viola-Jones) on both metrics — it has higher precision (85.7% vs 80.0%), meaning fewer of its positive detections are false alarms, and higher recall (81.8% vs 72.7%), meaning it misses fewer actual objects. Since Model B dominates on both precision and recall simultaneously, it provides better overall object detection performance without any trade-off between the two metrics.

Justification: A higher precision reduces false alarms (important to avoid unnecessary alerts), while a higher recall reduces missed detections (important for safety and completeness). Because Model B achieves superior values on both fronts, it is the clearly better-performing model for this detection task.

Q2. Motion Estimation Error Analysis

Given Data (Displacement Errors in Pixels)

Method	Frame 1	Frame 2	Frame 3
Method A	2	3	2
Method B	3	5	4

(a) Average Displacement Error

$$\text{Average Error} = (\text{Sum of errors}) / (\text{Number of frames})$$

Method A: $(2 + 3 + 2) / 3 = 7 / 3 = 2.33$ pixels

Method B: $(3 + 5 + 4) / 3 = 12 / 3 = 4.00$ pixels

(b) & (c) Comparison and Justification

Method A has a lower average displacement error (2.33 pixels) compared to Method B (4.00 pixels). A lower error indicates that Method A's estimated motion vectors are closer to the true displacement in each frame. Therefore, Method A provides more accurate motion estimation than Method B.

Justification: Since motion estimation accuracy is judged by how closely the estimated displacement matches the actual displacement, and Method A consistently records smaller errors across all three frames (2, 3, 2 vs 3, 5, 4), it is the more reliable and precise technique for tracking motion between consecutive frames.

Q3. Depth Estimation Comparison

Given Data

Object	Actual Depth (m)	Stereo Vision (m)	SfM (m)
Object 1	2.0	2.0	2.5
Object 2	3.0	3.0	3.8
Object 3	4.0	4.0	5.0

(a) Absolute Error per Object

$$\text{Absolute Error} = | \text{Estimated Depth} - \text{Actual Depth} |$$

Stereo Vision: Object 1 = $|2.0 - 2.0| = 0.0$ m; Object 2 = $|3.0 - 3.0| = 0.0$ m; Object 3 = $|4.0 - 4.0| = 0.0$ m

SfM: Object 1 = $|2.5 - 2.0| = 0.5$ m; Object 2 = $|3.8 - 3.0| = 0.8$ m; Object 3 = $|5.0 - 4.0| = 1.0$ m

(b) Average Absolute Error

Stereo Vision: $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m

SfM: $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.767$ m

Method	Avg. Absolute Error
Stereo Vision	0.0 m
Structure-from-Motion (SfM)	0.767 m

(c) Comparison and Evaluation

Stereo Vision produces exact matches with the actual depth values (zero error across all three objects), while SfM shows a consistent overestimation of depth, with error increasing at greater distances (0.5 m, 0.8 m, 1.0 m). This indicates Stereo Vision provides significantly more accurate depth estimation for this dataset.

Justification: Since accuracy in depth estimation is measured by closeness to ground truth, and Stereo Vision's average absolute error (0.0 m) is far lower than SfM's (0.767 m), Stereo Vision is the more accurate method here. SfM's error growth with distance also suggests it may be less reliable for longer-range depth estimation without additional calibration or scale correction.